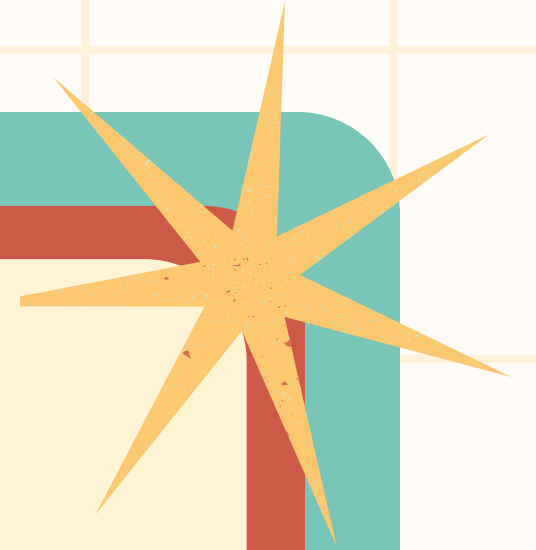


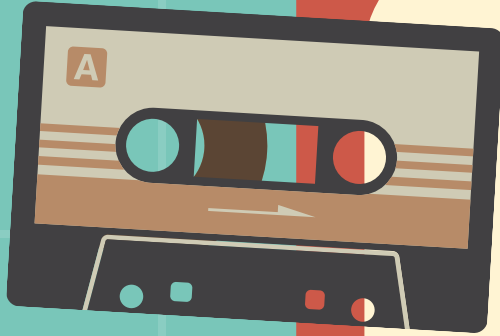
DATA STORYTELLING PROJECT

● SHANZA ASIF ●



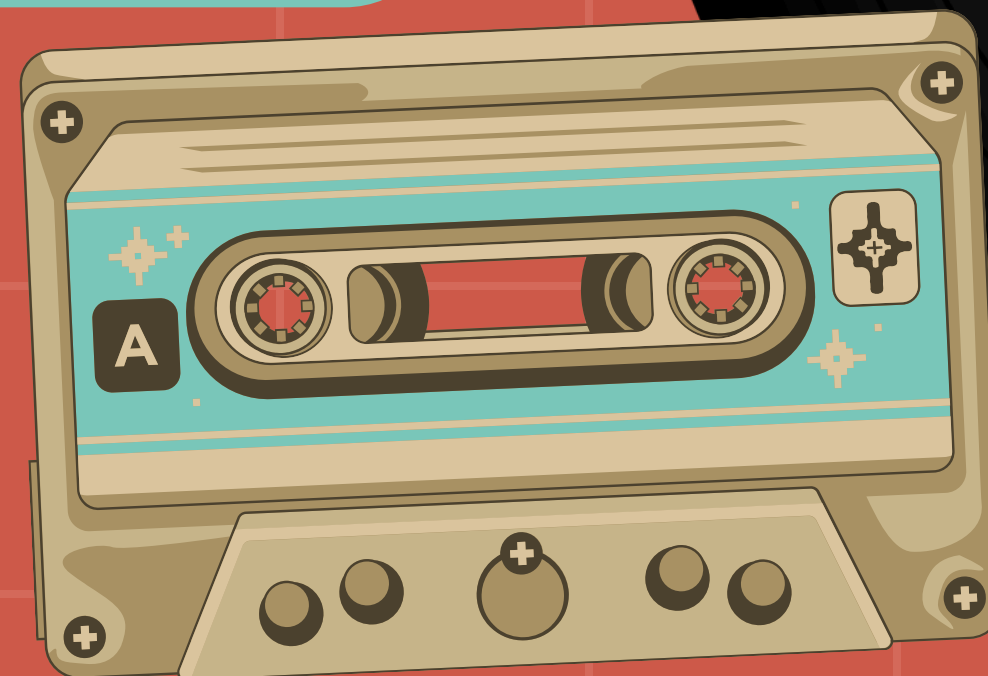
CONTEXT

- Spotify has over 600 million users worldwide, millions of them across Europe
- Every country has its own culture, language, and history
- Music reflects emotion, identity, and how people feel



RESEARCH QUESTION

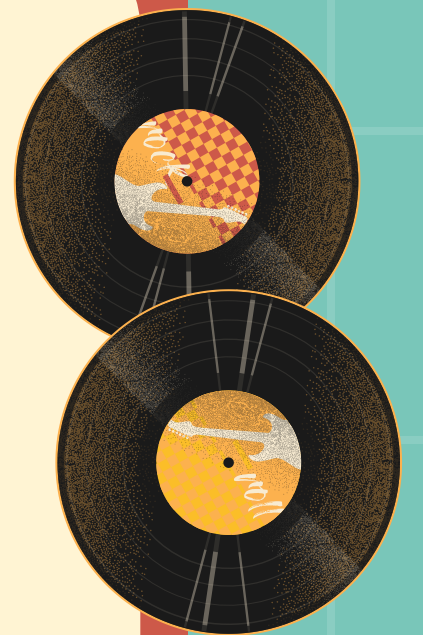
Do European countries have distinct musical moods, and do geographic patterns emerge from the data?



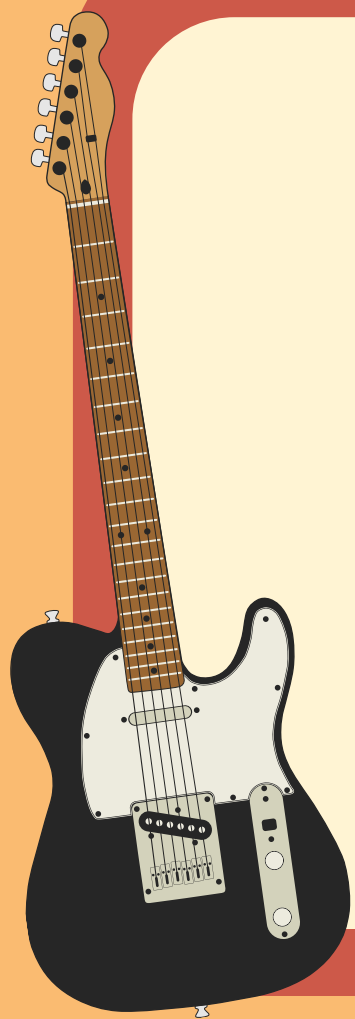
DATA

- Spotify Charts (All Audio Data) by Sunny Kakar from Kaggle
- Spotify's top 200 chart, across 69 countries worldwide

- Filtered to 27 European countries
- Date range: 2017 and 2018
- Each entry includes: track name, artist, rank, streams, and Spotify audio features



AUDIO FEATURES



Energy

intensity and activity of a song (0 to 1)

Valence

musical happiness of a song (0 to 1)

CLEANING

- Dataset covered 2017, 2018, 2019, 2020, and 2021
- 2019 had only 6 rows for all of Europe: unusable
- 2021 had only 28 rows: unusable

- 2020 had 41,000 rows, but only covered May and June
- Decision: keep only 2017 (172,000 rows) and 2018 (93,000 rows)

SCORING

- Equal weighting of energy and valence reflects equal contribution to musical mood

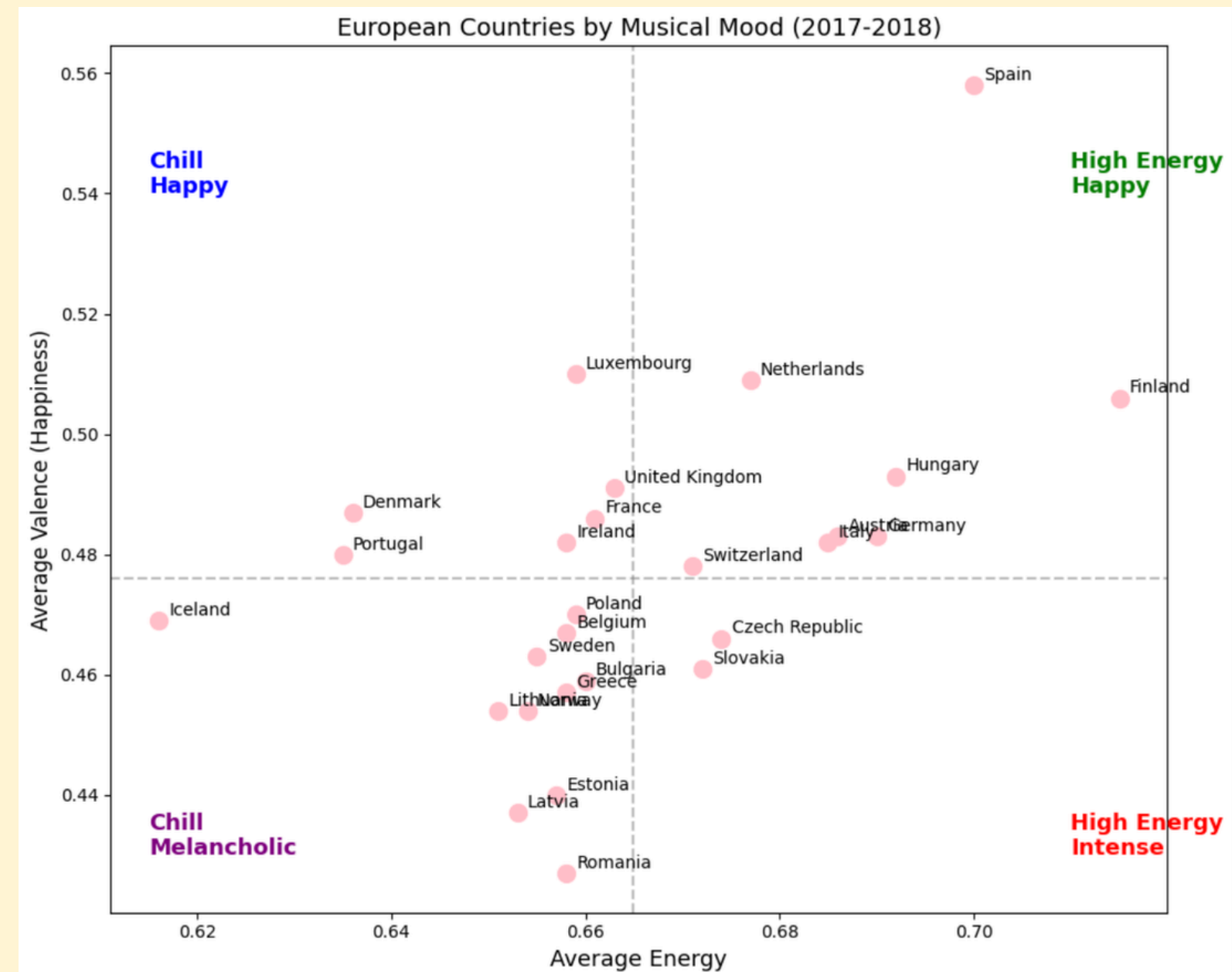
	af_energy	af_valence	mood_score
region			
Spain	0.700	0.558	0.629
Finland	0.715	0.506	0.610
Netherlands	0.677	0.509	0.593

- Higher score = more upbeat, Lower score = more melancholic.

EUROPE'S MUSICAL MOOD LANDSCAPE

- Grouped all chart entries by country and calculated the mean energy and valence across all songs and both years
- Split into 4 quadrants
- Labeled each dot with the country name so patterns are immediately readable

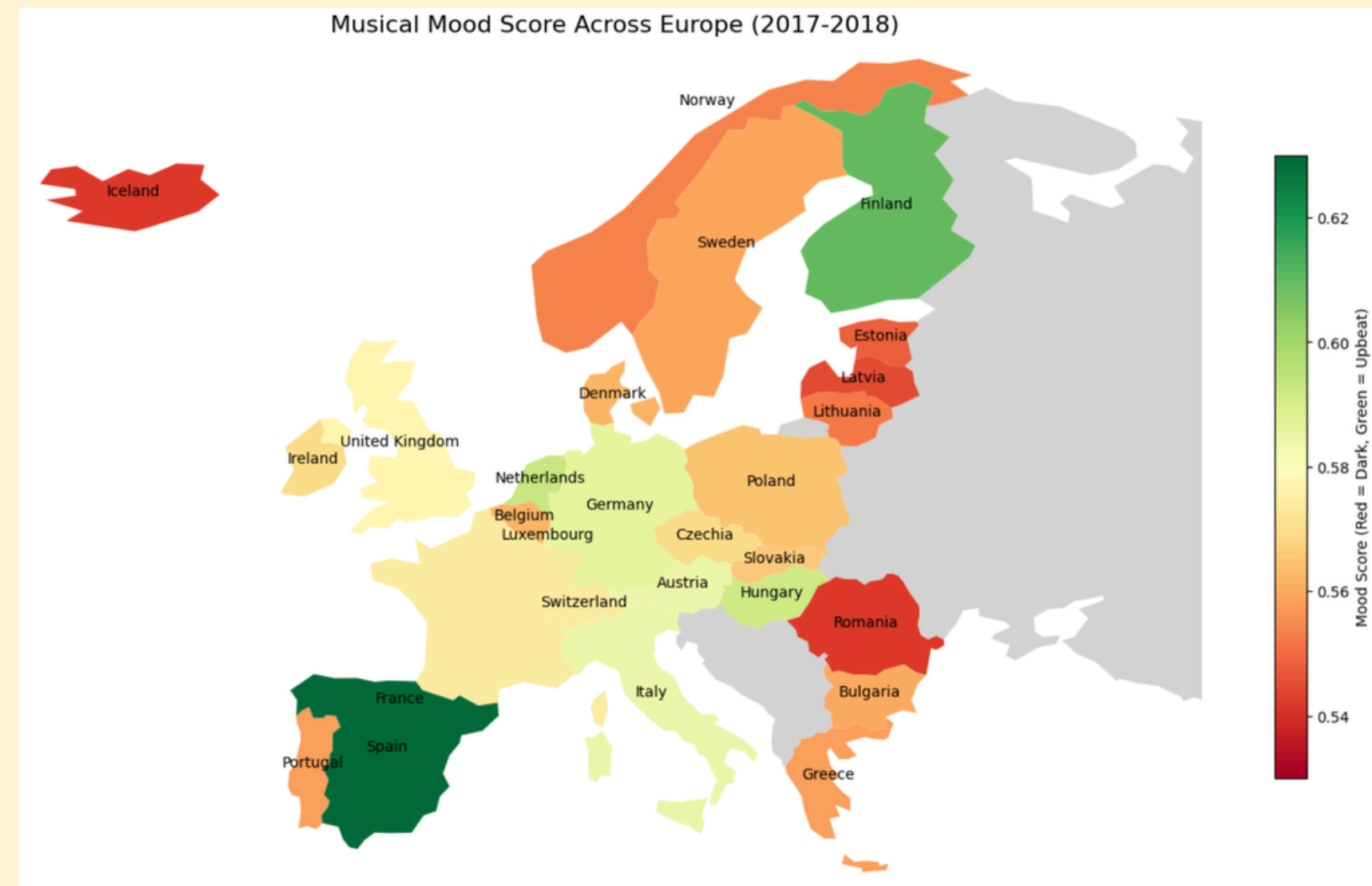
- Spain: outlier – uniquely high energy AND happy,
- Iceland and the Baltic cluster, bottom left – the most melancholic
- Most of Western Europe in the Middle



GEOGRAPHY AND MUSICAL MOOD ARE CONNECTED

- Calculated the mood score for each country,
- Merged data with a geographic shapefile of Europe using the geopandas library
- Applied a color scale tied to the mood score values

- Spain and Finland – the highest mood scores in Europe
- Romania, the Baltics, and Iceland – lowest mood scores
- Central and Western Europe - middle ground



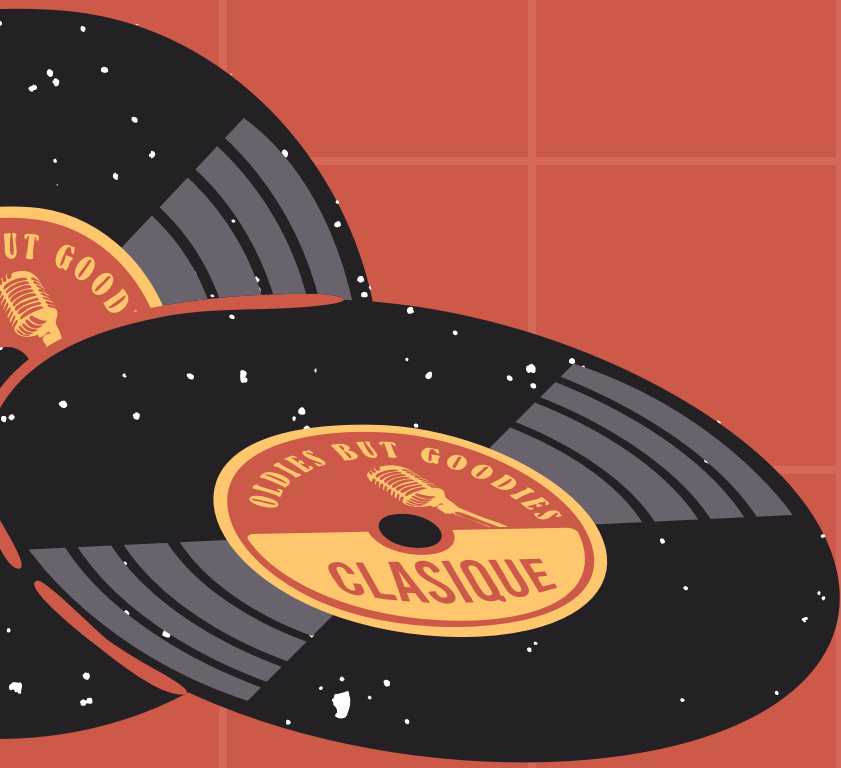
LIMITATIONS

- Data only captures songs that made the daily Top 200.
- 2019 and 2021 had fewer than 30 rows of European data and were dropped. 2020 only covered May and June, so a COVID analysis wasn't possible
- Germany has 83 million people with vastly different tastes. One average number doesn't capture that complexity





TAKEAWAYS



- European countries have distinct musical moods
- A clear geographic pattern
 - Southern Europe trends more upbeat, Eastern Europe and the Nordic area trends more melancholic
- Spain is the biggest outlier!
 - Happier and more energetic than every other country
- The Baltics (Estonia, Latvia, Lithuania) consistently score lowest for valence

Music is a measurable reflection of culture!!

THANK
• YOU •

